

Exploratory Experiments — Team Report

Project: Enhancing Loan Default Prediction Using Topological Data Analysis

Audience: Research team

Scope: Exploratory experiments that are **not** in the main paper tables, but that shaped what is considered sensible for this project.

Excluded from this report (out of project scope): Experiment 20 (deep learning placeholder); Experiment 5 Statlog Mapper stubs.

Purpose of this report

The paper-facing experiments (1–4, 6, 11–14, 19) answer predictive-performance questions. The exploratory set answers a different question: **what topological and geometric structure is present, and which modelling choices are defensible?**

They form the basis for:

- choosing Mapper / visualisation parameters,
- deciding whether barcode features are redundant or geometrically separated,
- stress-testing PCA dimension and KNN neighbourhood size,
- motivating the statistical follow-ups (Experiments 24–27) and the early-split protocol (Experiment 23).

Experiment map

| Exp | Folder | Role in the research narrative | Status |

|----|-----|-----|-----|

| 5 | `5\_Mapper` | Shape of **original** features via Kepler Mapper | DCCCD implemented; Statlog stubs ignored |

| 7 | `7\_EDA\_Barcode\_Statistics` | EDA of barcode matrices from Exp 3/6 | Complete artefacts in results |

| 8 | `8\_Dimensionality\_Reduction\_On\_Barcode\_Statistics` | PCA view of barcode space | Complete |

| 9 | `9\_Dimensionality\_Reduction\_On\_Original\_Dataset` | PCA view of original processed data | Complete |

| 10 | `10\_Covariance\_Matrix\_And\_Distances` | Centroid / distance geometry of barcode clouds | Complete |

| 15 | `15\_Working\_With\_K\_in\_KNN` | Sensitivity of KNN to `k` on barcodes | Complete |

| 16 | `16\_Variance\_Retained\_...\_Default\_...` | PCA component sweep (DCCCD) | Complete |

| 17 | `17\_Distribution\_For\_Each\_Class` | PCA / t-SNE / UMAP class separability | Complete |

| 18 | `18\_Variance\_Retained\_...\_Statlog\_...` | PCA component sweep (SGCD) | Complete |

| 21 | `21\_Visualizing\_Data\_Shape\_...\_Using\_TDA` | Mapper on **barcode** statistics | Complete (HTML outputs) |

| 22 | `22\_Visualizing\_Persistence\_Diagrams` | Persistence diagrams per class | Present; verify Statlog path |

Findings by experiment

Experiment 5 — Mapper on original data (DCCCD)

Goal. Explore whether default vs non-default customers form connected structure under filter functions (PCA lens) and cover parameters (resolution, overlap, clustering).

Method. Kepler Mapper grids over resolution / overlap / KMeans `k`, producing interactive HTML graphs under `5\_Experiments/5\_Mapper/...` and mirrored results.

Why it matters. Supports the claim that topology is a *plausible* lens on credit data before committing to PH + ML. It does **not** produce a production classifier; it informs intuition and figure design.

Caveat. Large HTML volume; Statlog variants were not finished and are out of scope here.

Experiment 7 — EDA on barcode statistics

Goal. Characterise the barcode-statistic feature space (H■+H■ and H■-only) after Exp 3 / Exp 6.

Method. `eda(..., graphs=True)` on per-class and full barcode CSVs; stored as pickles under `6\_Results/7\_EDA\_Barcode\_Statistics/`.

Why it matters. Shows which barcode columns are skewed, sparse, or class-associated before we trust ML metrics. Feeds Experiment 11 (correlation dropping) conceptually.

Experiments 8 & 9 — PCA on barcodes vs original data

Goal. Compare low-dimensional geometry of barcode features (Exp 8) with original processed features (Exp 9).

Method. `perform\_pca\_analysis` → scree plots, 2D scatters, loadings CSVs.

Why it matters. If classes separate more cleanly in barcode PCA than in original PCA (or vice versa), that guides interpretation of TDA gains. Also justifies later matched-variance experiments (13, 16, 18).

Experiment 10 — Covariance and centroid distances

Goal. Probe whether barcode clouds for the two classes are geometrically distinguishable via mean / farthest / random centroids.

Method. Distance matrices and centroid-based summaries saved under `6\_Results/10\_Covariance\_Matrix\_And\_Distances/` (L5/L15 and L30/L60).

Why it matters. Complements black-box ML scores with a geometric story: if centroids are far in barcode space, topological summaries carry class signal.

Experiment 15 — K in KNN

Goal. Elbow / sensitivity analysis for KNN on barcode features (`k = 1...20`).

Method. `train_multiple_knn_datasets` + elbow curve plots.`

Why it matters. Guards against over-interpreting a single default `k``. Informs whether neighbourhood methods are stable in barcode space.

Experiments 16 & 18 — PCA component sweeps

Goal. Vary the number of PCA components used before PH and retrain, for DCCCD (16) and SGCD (18).

Method. `run_experiments_with_pca_components`` over a grid (e.g. 2...19) with metric-vs-components plots.

Why it matters. PCA-7 / PCA-15 were modelling choices, not theorems. These sweeps show sensitivity of downstream accuracy to that choice and motivate Experiment 26 (estimate intrinsic dimension `b`` properly).

Experiment 17 — Class distribution visualisations

Goal. 2D/3D PCA, t-SNE, UMAP of barcode features with optional animations.

Method. `visualize_class_separability``.

Why it matters. Qualitative evidence for separability (or overlap) that numbers alone miss. Useful for slides and thesis figures.

Experiment 21 — Mapper on barcode statistics

Goal. Apply Mapper to the **TDA** feature space rather than original attributes.

Method. `build_mapper_viz`` with PCA/UMAP lenses on `data_L*` matrices.

Why it matters. Closes the loop: if Mapper structure appears in barcode space, our PH summaries preserve organised geometry—not only tabular signal.

Experiment 22 — Persistence diagrams

Goal. Plot persistence diagrams for default vs non-default point clouds (Ripser + persim).

Method. Direct diagram visualisation (not barcode aggregation).

Why it matters. Pedagogical and diagnostic: confirms that PH is computing non-trivial structure.

Action item: verify the Statlog script is not accidentally loading DCCCD paths.

How these feed the next experiments

| Exploratory insight | Follow-up |

|-----|-----|

| Full-data PCA/landmarks may leak | **Exp 23** early 80/20 + independent train/test snapshots |

| `l=500`` and large `t`` look aggressive | **Exp 24** sampling-ratio audit |

Need mean/variance of snapshots for theory	**Exp 25** snapshot statistics / $\bar{\lambda}$ proxy
PCA dims $\neq$ intrinsic dimension	**Exp 26** Two-NN & Levina–Bickel
Need formal two-sample evidence	**Exp 27** Algorithm 2 / $F_{\{p,q\}}$ (Robinson & Turner)

Recommended team actions

1. Read `docs/Pipeline\_Issues\_And\_Leakage.md` alongside this report.
2. Review CV numbers in `docs/CV\_Results.md` (paper experiments).
3. Run Exp **24** and **26** first (fast feedback on ratios and b).
4. Run Exp **25** / **27** on existing Exp-3 barcode CSVs.
5. Schedule Exp **23** when compute time for landmarks is available.
6. Cite Robinson & Turner (arXiv:1310.7467) and Chazal et al. (arXiv:1406.1901) in the write-up; reference Frontiers TDA survey §6.3.1 for snapshot statistics.

References (for the statistical thread)

- Robinson, A. & Turner, K. *Hypothesis Testing for Topological Data Analysis*. arXiv:1310.7467 (Algorithm 2, loss $F_{\{p,q\}}$).
- Chazal, F. et al. *Subsampling Methods for Persistent Homology*. arXiv:1406.1901 ($\bar{\lambda}$, bias/variance of subsample landscapes).
- Frontiers survey: *An Introduction to Topological Data Analysis* — Section 6.3.1 (snapshot mean/variance; predictive use of estimators).
- Facco et al. — Two-NN intrinsic dimension.
- Levina & Bickel — MLE intrinsic dimension.

Generated for internal team circulation. Update after Exp 23–27 results land in `6\_Results/`.